You'll use many AWS services to implement the Imagery application. Most of them are fault tolerant by default, but EC2 isn't. You'll deal with that problem using an idempotent state machine, as introduced in the next section.

Example is 100% covered by the Free Tier

The example in this chapter is totally covered by the Free Tier. As long as you don't run the example longer than a few days, you won't pay anything for it. Keep in mind that this applies only if you created a fresh AWS account for this book and there is nothing else going on in your AWS account. Try to complete the example within a few days, because you'll clean up your account at the end of the section.

16.3.1 The idempotent state machine

An idempotent state machine sounds complicated. We'll take some time to explain it because it's the heart of the Imagery application. Let's look at what a *state and machine* is and what idempotent means in this context.

The finite state machine

A finite state machine has at least one start state and one end state. Between the start and the end states, the state machine can have many other states. The machine also defines transitions between states. For example, a state machine with three states could look like this:

```
(A) -> (B) -> (C).
```